

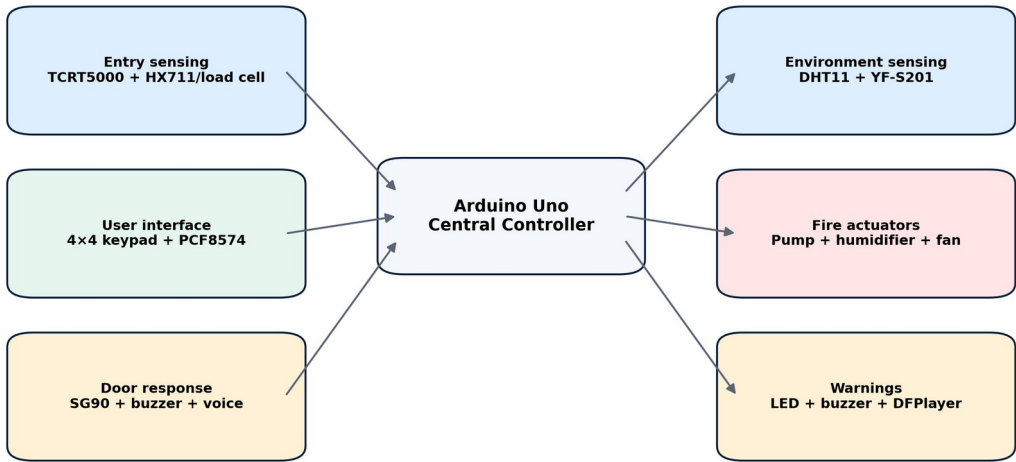
LEBANESE UNIVERSITY

Faculty of Engineering III | Department of Computer and Communications Engineering
Sensors & Instrumentation

Security and Fire-Protection System
FINAL TECHNICAL REPORT



Smart Biological Laboratory Security and Fire-Protection Architecture



Two coordinated subsystems share one controller while preserving independent decision logic.

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This project delivers an Arduino-based smart sensing platform for a building. The system combines controlled-entry security with automatic fire detection and environmental response. Five required sensing modalities are integrated: optical detection, applied force/weight, liquid flow, temperature, and humidity. The design acquires sensor data, performs threshold-based processing, controls electromechanical actuators, logs timestamped CSV records, and provides audible and visual feedback.

The entry subsystem uses a TCRT5000 infrared sensor and an HX711-based load cell as a dual-confirmation gate. A PIN is requested only when optical presence and entry-platform weight are both confirmed. Correct credentials open a servo-controlled door for five seconds; three incorrect attempts activate a latched security alarm. The fire subsystem uses DHT11 temperature and humidity readings, a YF-S201 flow sensor, a pump, an ultrasonic humidifier, and a ventilation fan. The controller operates through a state machine with fire detection, pumping, blockage monitoring, humidification, and recovery states.

Experimental evidence at a glance

The organized test campaign contains 242 multi-sensor samples, 113 system events, three acquisition sessions, a longest continuous run of 1,168.41 s, four fire cycles (three complete), and 15 physical load-cell calibration trials. The main integrated run satisfies the minimum 60-second continuous-operation requirement by a wide margin.

Metric	Result	Interpretation
Samples	242	Multi-sensor snapshots
Longest continuous run	1,168.41 s	Approximately 19.47 min
Median main-run interval	5.018 s	Matches the 5 s firmware target
Event records	113	Security, fire and firmware events
Complete fire cycles	3 of 4	Fourth cycle was still pumping at end of capture
Grade-A samples	211 (87.2%)	High overall dataset quality
Load-cell corrected MAPE	98.60%	Scale factor must be recalibrated

Key engineering conclusion

The overall architecture and automatic responses are functional, but three issues require closure before final deployment: the load-cell scale factor is invalid, the fan threshold values are inconsistent between documentation and extracted constants, and the 24 V humidifier exceeds the course specification of 12 V maximum unless instructor approval and electrical isolation are formally documented.

B

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Symbols and abbreviations

Term	Meaning	Term	Meaning
ADC	Analog-to-digital count	Q	Flow rate, L/min
f	Pulse frequency, pulses/s	RH	Relative humidity, %
MAPE	Mean absolute percentage error	CSV	Comma-separated values
I2C	Inter-integrated circuit bus	GPIO	General-purpose I/O
I/O	Inputs, outputs and relay contacts		

1

Introduction

Project purpose, context and contribution

The building contains controlled-access zones, sensitive equipment, electrical loads, chemical materials, and environmental risks. A conventional installation often treats access control, fire detection, ventilation, alarms, and data logging as separate systems. The proposed prototype demonstrates how heterogeneous sensors can be integrated into a unified embedded platform without mixing the decision logic of the individual subsystems.

The project is implemented around an Arduino Uno and divides the application into two coordinated but independent subsystems. The entry-security subsystem protects the laboratory door. The fire-protection subsystem monitors environmental conditions and automatically activates suppression and ventilation hardware. A shared controller simplifies prototyping, while modular firmware and clear subsystem boundaries preserve maintainability.

1.1 Project objectives

- Integrate optical, force, flow, temperature and humidity sensing in one Arduino-based system.
- Acquire and log timestamped multi-sensor data with a nominal five-second reporting interval.
- Reduce false entry triggers by requiring simultaneous optical and load-cell confirmation.
- Detect abnormal temperature and control a pump, humidifier, fan, alarms and warning indicators automatically.
- Implement calibration, error analysis, data visualization, modular firmware and a reproducible GitHub structure.
- Demonstrate continuous operation for more than 60 seconds and explain all design decisions during the live demonstration.

1.2 Main contribution

The principal contribution is a compact laboratory monitor that links sensing, decision making, actuation, audio guidance, and data engineering. The load cell is deliberately assigned only to the entry platform; it is not used as a water-tank sensor. This corrected allocation gives each required sensor modality a distinct and meaningful engineering role.

2

Problem Statement

Why a dual-purpose laboratory monitor is needed

A laboratory entry system based on only one proximity sensor may trigger because of reflections, nearby objects, or ambient-light changes. Similarly, a fire alarm that only reports high temperature without checking water delivery cannot determine whether the suppression path is blocked. The problem is therefore not simply to read sensors, but to combine them into dependable decisions and automatic responses.

2.1 Functional problem

Hazard / need	Conventional weakness	Implemented response
Unauthorized access	Single-sensor triggers and repeated PIN attempts	Optical AND weight confirmation, PIN timeout, sticky thief alarm
Fire or overheating	Alarm without active response	Temperature threshold, warning LED, buzzer, pump and humidifier sequence
Blocked water path	Pump may run without useful delivery	Flow threshold and latched blockage state
Excess humidity	Humidity may remain high after suppression	Automatic ventilation fan control
Poor traceability	Manual observations are incomplete	Timestamped CSV logging and event records

2.2 Design specifications

Requirement	Target
Required modalities	Optical, force, flow, temperature, humidity

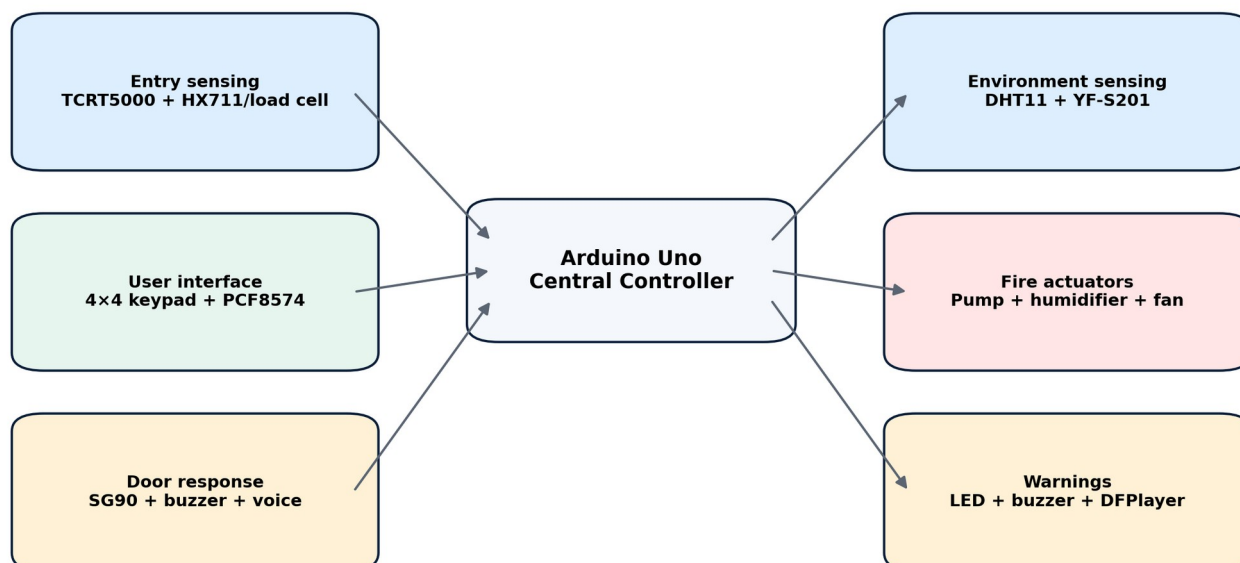
Requirement	Target
Entry confirmation	TCRT5000 detection AND entry weight ≥ 10 g
PIN inactivity timeout	15 s
Maximum wrong attempts	3
Door-open interval	5 s
Fire threshold	28 °C target constant
Flow blockage threshold	0.05 L/min
Pump duration	55 s nominal
Humidifier interval	10 s measured timer window / RH target check
Logging interval	5 s
Continuous demonstration	At least 60 s

3

System Architecture

Subsystem boundaries, data flow and control

Smart Biological Laboratory Security and Fire-Protection Architecture



Two coordinated subsystems share one controller while preserving independent decision logic.

Figure 3.1 - Functional block architecture

The Arduino Uno coordinates all sensing and actuation. Entry-security tasks are polled continuously for rapid response. Fire-protection processing and CSV output execute every five seconds. The security and fire state variables remain separate: entry weight cannot initiate, stop, or modify the fire sequence.

3.1 Entry-security data path

```

opticalDetected = (TCRT5000 ADC < 500)
weightDetected = (entry-platform weight >= 10 g)
personDetected = opticalDetected AND weightDetected

```

When personDetected becomes true and the debounce interval has elapsed, the DFPlayer prompts the user for a PIN. The keypad is scanned through the PCF8574 expander. A correct PIN resets the alarm and rotates the SG90 servo to open the door. An incorrect PIN increments the attempt counter; after three failures the buzzer remains active until a correct PIN is entered.

3.2 Fire-protection data path

Fire-Protection State Machine

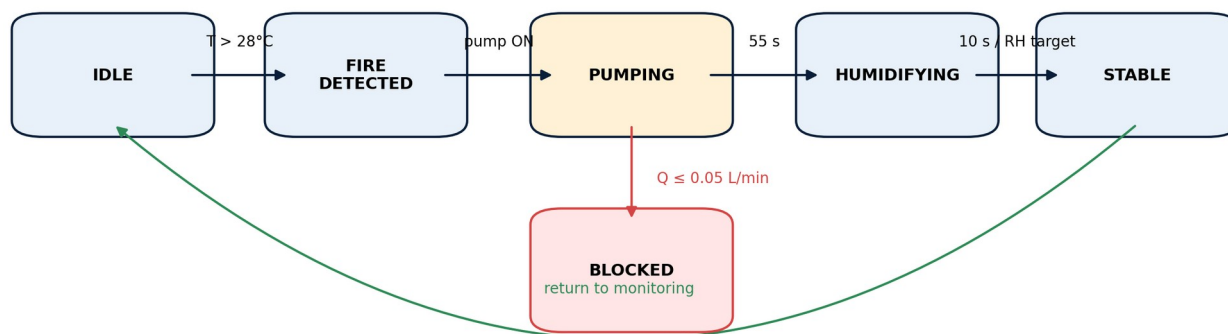


Figure 3.2 - Fire-protection state machine

The DHT11 provides temperature and humidity, while the YF-S201 produces flow pulses. High temperature moves the controller from IDLE to FIRE_DETECTED, activates warning outputs, and starts the pump. During pumping, the measured flow is compared with the 0.05 L/min blockage threshold. After the pump interval, the humidifier is activated and the system returns to stable monitoring after the recovery condition.

3.3 Data acquisition and visualization

The serial stream contains timestamp, temperature, humidity, entry weight, optical state, weight state, combined person state, water flow, fire stage, and wrong-attempt count. This format can be imported directly into Excel or a live serial plotter. The main experimental session used a median interval of 5.018 s, closely matching the firmware target of 5 s.

Role	Component	Measured / controlled quantity	Interface	Justification
Optical	TCRT5000 IR module	Object/person presence	Analog A2	Low-cost short-range reflective detection; threshold can be tuned for the installed distance.
Force	Load cell + HX711	Applied entry-platform weight	Digital serial interface	Adds physical confirmation and rejects optical-only false triggers.
Flow	YF-S201	Water flow, L/min	Digital interrupt D2	Pulse output supports blockage detection and delivered-volume estimation.
Temperature	DHT11	Ambient temperature, °C	Digital D8	Direct input to abnormal-temperature detection.
Humidity	DHT11	Relative humidity, %RH	Digital D8	Controls recovery and ventilation after suppression.
User input	4x4 keypad + PCF8574	PIN digits	I2C, address 0x20	Reduces Arduino pin consumption and supports clear access control.
Door actuator	SG90 servo	Door angle	PWM control	Provides repeatable open/close motion.
Audio	DFPlayer Mini + speaker	Voice prompts and warnings	Serial control	Improves usability and communicates system state without a display.

4.1 Distinct role of the load cell

Architecture rule

The load cell is an entry-security sensor only. It measures applied weight on the entry platform and must never be described as a tank-level or fire-protection sensor.

4.2 Signal-conditioning considerations

- The HX711 provides amplification and high-resolution conversion for the load cell; tare is executed with the platform empty.
- The TCRT5000 threshold is applied to a raw ADC reading and must be tuned under the laboratory lighting and mounting distance.
- The YF-S201 pulse line is connected to an interrupt-capable input; pulse frequency is converted to L/min.
- Relay modules isolate Arduino logic from the pump, humidifier and fan load currents. Active-low logic requires a defined OFF state during startup.
- The warning LED requires a current-limiting resistor, and all modules share an appropriate reference ground on the low-voltage control side.

5

Hardware Design and Schematics

Prototype wiring, power domains and physical integration

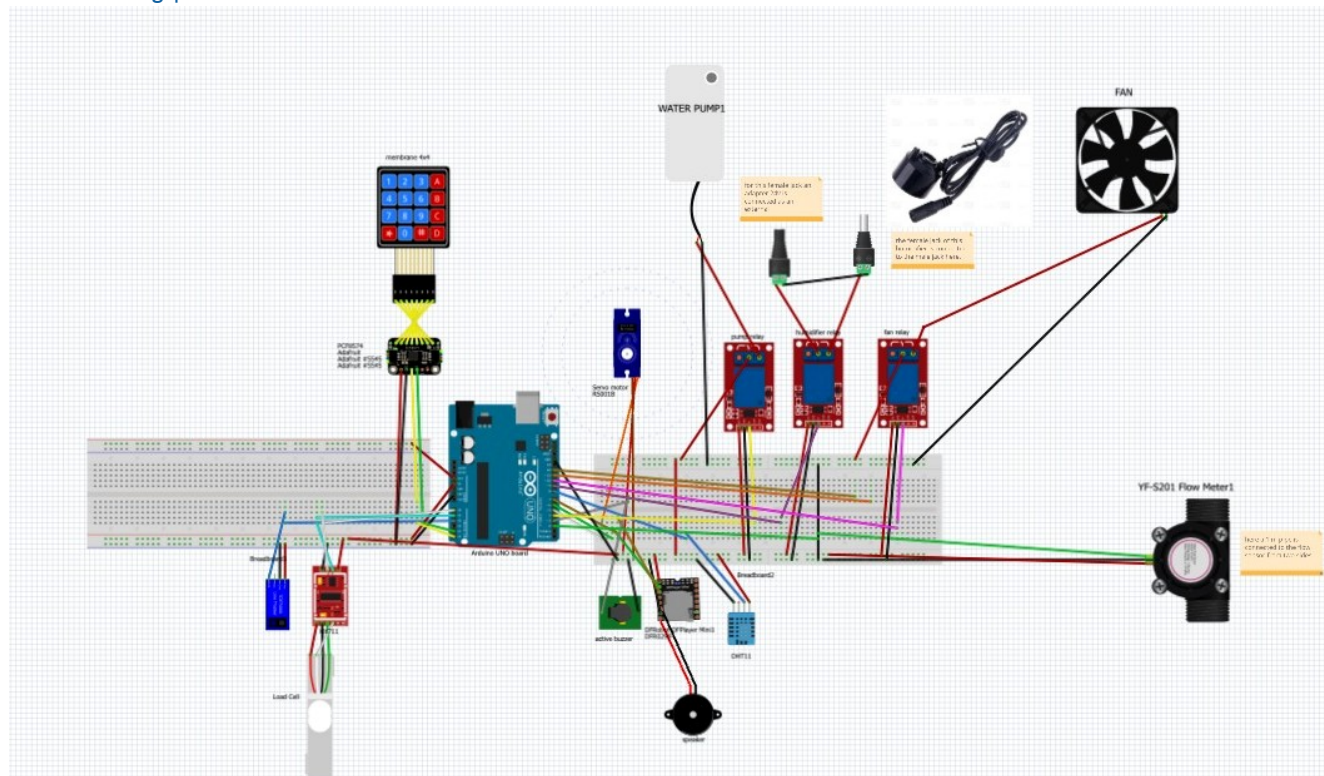


Figure 5.1 - Final Fritzing breadboard representation of the integrated prototype

The breadboard layout places the Arduino Uno at the center, with the entry hardware on the left and environmental actuators on the right. Three active-low relay modules switch the pump, humidifier and fan. The keypad is routed through the PCF8574 expander, and the audio subsystem is positioned near the DHT11 and controller. Male and female barrel connectors represent the external humidifier power connection.

5.1 Known interface map

Device	Verified interface	Notes
TCRT5000	A2	Raw ADC compared with the optical threshold
DHT11	D8	Temperature and humidity, read every 5 s
YF-S201	D2 interrupt	Pulse counting for flow conversion
PCF8574 keypad expander	I2C address 0x20	Rows and columns connected through the expander
SG90 servo	PWM-capable digital output	0° closed, 180° open in corrected algorithm
Pump / humidifier / fan relays	Active-low digital outputs	COM/NO contacts switch the load supply
DFPlayer Mini	Serial interface	Voice prompts and warnings

5.2 Power and protection

- Arduino and logic modules operate from the regulated low-voltage control supply.
- The pump, fan and humidifier are powered through relay contacts rather than Arduino GPIO pins.
- The 24 V humidifier adapter is connected only on the isolated relay-load side; the Arduino must never receive 24 V.
- Barrel-jack polarity is center-positive unless the actual adapter label specifies otherwise.

- Inductive loads should include appropriate flyback protection where it is not already provided by the relay module or load driver.
- Water tubing and the electronics should be physically separated to reduce leak risk.

Course-voltage compliance

The course specification states a maximum operating voltage of 12 V DC. The current 24 V humidifier therefore requires formal instructor approval or replacement by a 12 V unit. Electrical isolation alone does not remove this specification conflict.

5.3 Water path

The recommended hydraulic path is: water container -> mini pump -> YF-S201 flow sensor -> humidifier reservoir or discharge outlet. Six-millimeter flexible tubing, secure fittings and leak testing are required. The flow sensor should be oriented according to its arrow and kept full of water during measurement to reduce air-bubble error.

6

Software Design and Firmware Architecture

Modular control, timing and fault handling

The firmware is organized into a main Arduino file and subsystem modules for entry security and fire protection. Startup configures all pins, initializes peripherals, tares the load cell, sets active-low relays OFF, closes the servo door and prints the CSV header.

6.1 Main scheduler

Task	Execution rate	Reason
Optical and load-cell polling	Continuous	Fast reaction at the door
Keypad processing	During active PIN session	Immediate feedback and timeout restart
Non-blocking buzzer update	Continuous	Alarm timing without stopping other tasks
DHT11 and flow processing	Every 5 s	Sensor limitations and manageable log rate
Fire-state update	Every 5 s	Coordinated state progression
CSV record output	Every 5 s	Analysis-ready time series

6.2 Entry algorithm

1. Read the TCRT5000 ADC value and the most recent HX711 weight.
2. Calculate opticalDetected and weightDetected using the configured thresholds.
3. Require the logical AND of both conditions before requesting a PIN.
4. Accept numeric keys, use # to submit, * to clear, and ignore A-D.
5. Open the door for five seconds after a correct PIN.
6. After three incorrect attempts, latch the security alarm until a correct PIN is entered.

6.3 Fire algorithm

1. Read temperature, humidity and accumulated flow pulses.
2. When temperature exceeds the fire threshold, activate the warning outputs and enter FIRE_DETECTED.
3. Start the pump and monitor flow. Enter BLOCKED when flow is at or below 0.05 L/min.
4. After the nominal 55-second pump interval, stop the pump and start humidification.
5. Stop humidification at the selected recovery criterion and return to stable monitoring.
6. Control the fan with separated ON and OFF thresholds to provide hysteresis.

6.4 Important firmware corrections

Issue observed	Engineering effect	Required correction
DHT failure skips the entire fire-state step	Pump completion may be delayed beyond 55 s	Check pump and alarm timers independently of DHT validity
Fan thresholds differ between algorithm text and extracted constants	Risk of chattering or contradictory behavior	Adopt one hysteresis pair, e.g. ON $\geq$ 80%RH and OFF $\leq$ 70%RH
Humidifier logic appears both timer-based and humidity-based	Documentation and test behavior may disagree	Use RH target with a 10 s maximum safety timeout
Load-cell factor 420 is an initial guess	Entry weight is severely inaccurate	Perform new raw-count calibration and store the final factor in a constants file or EEPROM

7

Calibration Procedures and Results

Three-point methods, equations and measured limitations

7.1 Calibration equations

$$\text{error} = \text{measured value} - \text{reference value}$$

$$\text{percentage error} = 100 \times \text{error} / \text{reference}$$

$$\text{MAPE} = \text{mean}(|\text{percentage error}|)$$

$$\text{linear correction: } y = m \times x + b$$

7.2 Physical load-cell calibration

Fifteen trials were recorded at three known reference weights: 500 g, 1,000 g and 2,000 g, with five repetitions per point. Signed readings were converted to positive magnitude because the mechanical mounting reversed the sign. The average measurements were 17.136 g, 6.199 g and 2.761 g respectively.

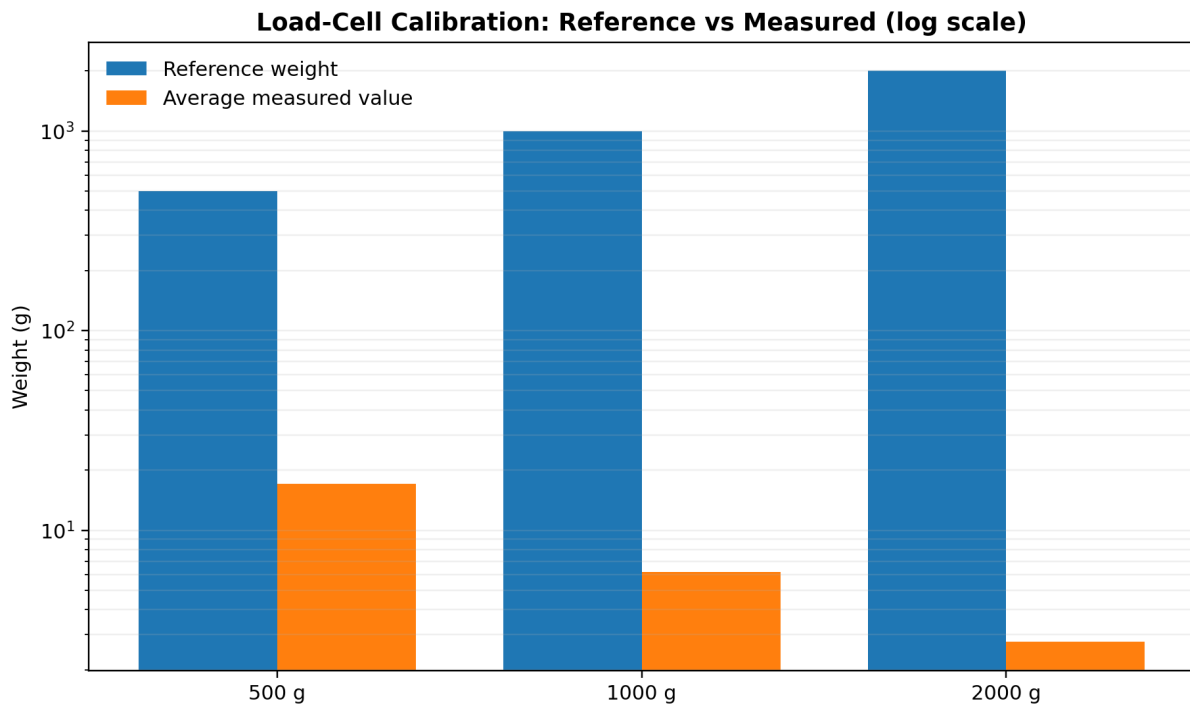


Figure 7.1 - Physical load-cell calibration results

Reference	Trials	Average measured	Average absolute error	MAPE
500 g	5	17.136 g	482.864 g	96.573%
1,000 g	5	6.199 g	993.801 g	99.380%
2,000 g	5	2.761 g	1,997.239 g	99.862%
Overall	15	-	-	98.605%

Calibration decision

The current scale factor cannot be accepted. The measured response is not proportional to the applied reference weight, so changing only the 10 g presence threshold would not solve the problem. Raw HX711 counts, tare offset and mounting polarity must be re-recorded before the final demonstration.

7.3 Required load-cell recalibration procedure

1. Unload the entry platform and execute tare.
2. Record the average raw HX711 count for at least 0 g, 500 g, 1,000 g and 2,000 g.
3. Fit reference weight against net raw count using linear regression.
4. Store the resulting scale factor and offset in the dedicated constants file or EEPROM.
5. Repeat each point at least five times and calculate bias, standard deviation and MAPE.
6. Choose the entry threshold only after the calibrated noise floor and intended platform load are known.

7.4 Statistical three-point validation for the remaining sensors

The environmental and optical datasets were also evaluated through three-point statistical correction. These points demonstrate the calculation method and sensitivity of a linear correction, but they are not a substitute for physical references such as a thermometer, hygrometer, controlled target distance or measured water volume.

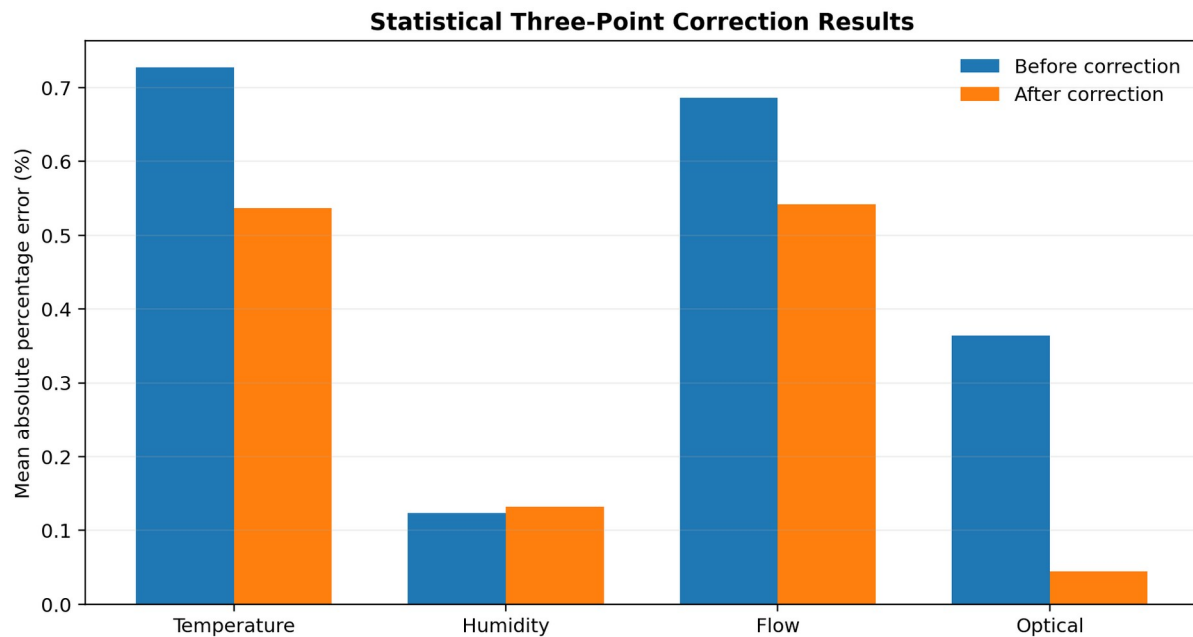


Figure 7.2 - Before/after error for statistical three-point correction

Sensor	Correction equation	Before MAE %	After MAE %	Interpretation
DHT11 temperature	$y = 0.8242x + 4.7802$	0.727	0.537	Small statistical improvement
DHT11 humidity	$y = 1.0021x - 0.1309$	0.123	0.132	No meaningful improvement
YF-S201 flow	$y = 1.0226x - 0.0210$	0.686	0.542	Small improvement
TCRT5000 optical	$y = 0.9674x + 29.5607$	0.364	0.045	Strong fit to selected statistical points

7.5 Physical three-point calibration still required

Sensor	Recommended physical references
DHT11 temperature	Three stable temperatures measured with a reference thermometer
DHT11 humidity	Three humidity conditions measured with a reference hygrometer
YF-S201	Three known collected volumes divided by stopwatch time
TCRT5000	Three controlled distances/targets under fixed lighting

8

Experimental Results and Data Analysis

Time series, events, automatic responses and calculated performance

8.1 Test campaign summary

Session	Samples	Start	End	Duration	Median interval
Session 1	11	5.207 s	153.318 s	148.111 s	15.536 s
Session 2	9	5.161 s	132.052 s	126.891 s	15.454 s
Main integrated run	222	5.107 s	1,173.517 s	1,168.410 s	5.018 s

The main run lasted approximately 19.47 minutes and therefore exceeded the required 60-second continuous demonstration by a factor of about 19.5. The median sampling interval of 5.018 s confirms that the primary logging schedule operated as intended.

8.2 Environmental measurements

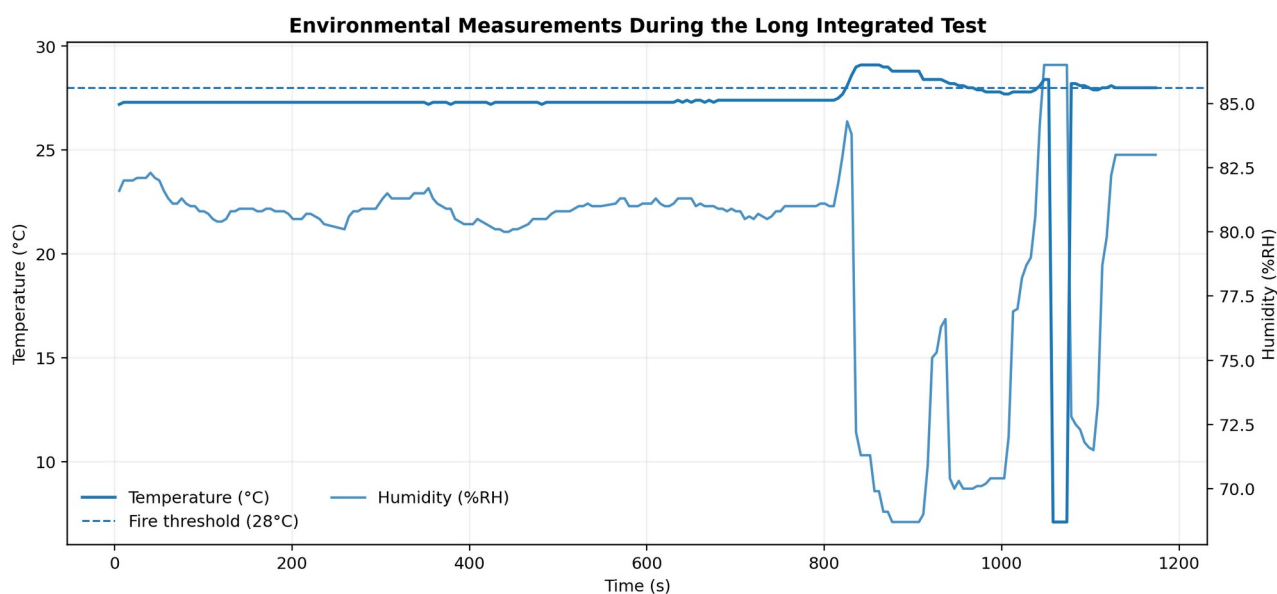


Figure 8.1 - Temperature and humidity during the main integrated run

Variable	Count	Minimum	Mean	Median	Maximum	Std. deviation
Temperature (°C)	242	7.10	26.82	27.30	29.10	3.01
Humidity (%RH)	242	68.70	78.64	80.80	86.50	4.79
Flow (L/min)	242	0.00	0.276	0.00	4.963	0.671

Thirty temperature values and thirty-one humidity values were marked as imputed or recovered, corresponding to 12.4% and 12.8% of the 242 samples. The event log contains 26 explicit DHT11 read-failure warnings. These failures did not stop the entire program, but the current implementation can delay state transitions because a failed read skips the fire-control step.

8.3 Water-flow response

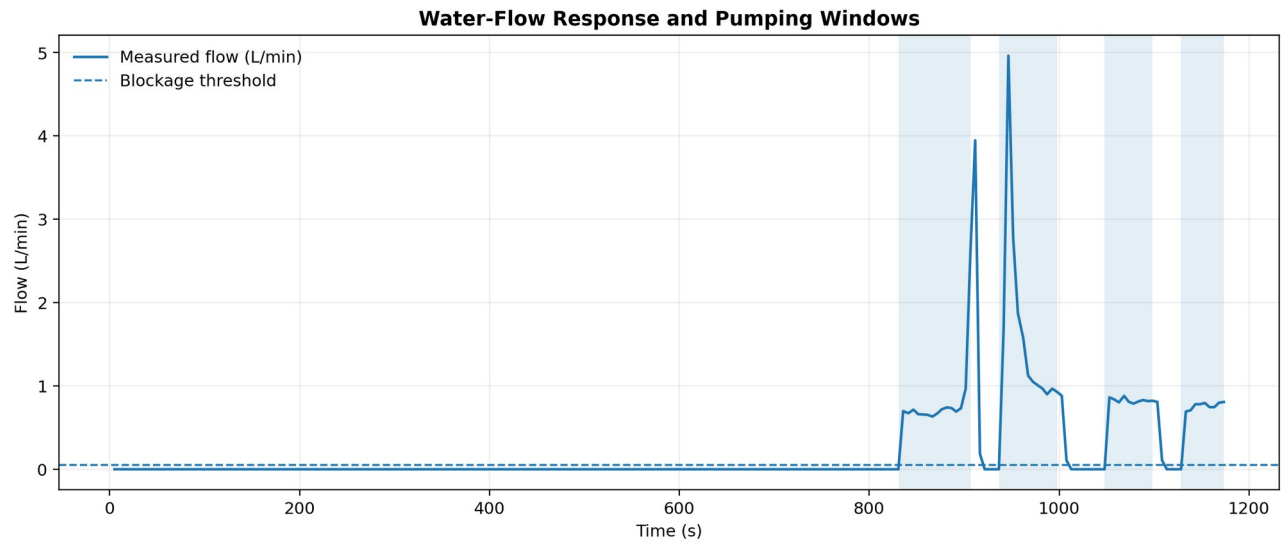


Figure 8.2 - Flow measurements and pumping intervals

$$Q \text{ (L/min)} = \text{pulse frequency (pulses/s)} / 7.5$$
$$\text{delivered volume (L)} = \text{integral of } Q \text{ dt} / 60$$

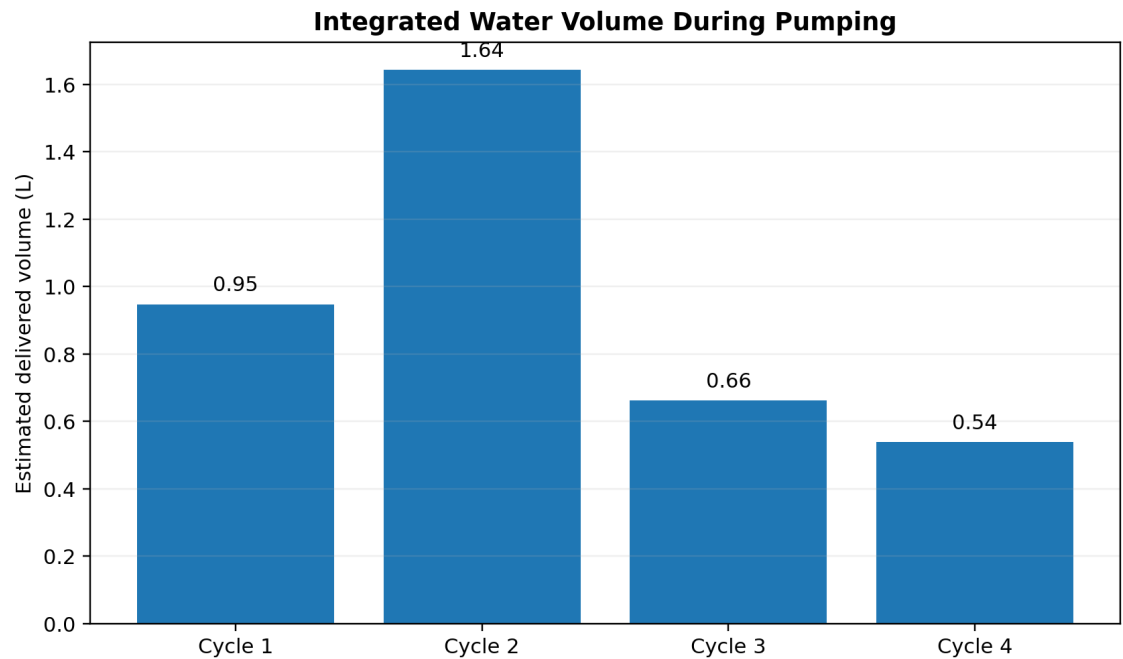


Figure 8.3 - Estimated volume delivered during each captured pumping window

Cycle	Pump-data duration	Mean flow	Maximum flow	Estimated volume	Status
1	75.881 s	0.785 L/min	2.614 L/min	0.948 L	Complete
2	61.233 s	1.527 L/min	4.963 L/min	1.643 L	Complete

Cycle	Pump-data duration	Mean flow	Maximum flow	Estimated volume	Status
3	50.515 s	0.751 L/min	0.881 L/min	0.661 L	Complete
4	45.043 s	0.686 L/min	0.807 L/min	0.538 L	Capture ended before completion

The integrated volumes are dataset-based estimates, not volumetric calibration standards. Each pumping cycle contained at least one sample at or below the 0.05 L/min blockage threshold, yet the recorded cycles generally continued. The final firmware should apply a short persistence window or startup grace period so that a single zero-flow sample at pump activation does not falsely latch BLOCKED.

8.4 Fire-cycle timing

Cycle	Fire detected	Pump complete	Observed interval	Humidifier interval
1	820.720 s	906.633 s	85.913 s	9.968 s
2	926.602 s	997.836 s	71.234 s	9.969 s
3	1,037.896 s	1,098.443 s	60.547 s	10.001 s
4	1,118.443 s	Not captured	Incomplete	Incomplete

The three completed humidifier intervals closely match ten seconds. The time from fire detection to pump completion exceeded 55 seconds, especially during cycle 1, because repeated DHT11 failures prevented the state update from checking the pump timer. This result directly supports moving timing checks outside the DHT-validity branch.

8.5 Security results

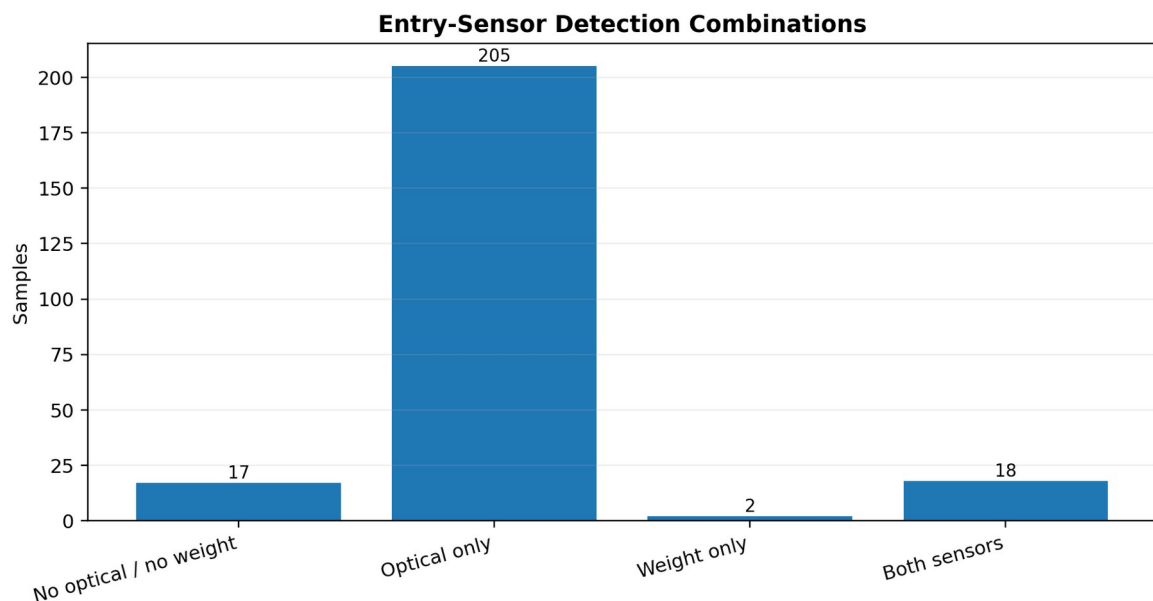


Figure 8.4 - Optical and load-cell detection combinations

At sample level, 18 records simultaneously satisfied optical and weight thresholds. Seven additional records were tagged as logged dual-sensor events even though the nearest sensor snapshot did not contain weight above 10 g; these are retained

as event-linked evidence rather than treated as threshold-valid samples. The event log also contains 19 legacy optical-only proximity triggers, demonstrating why the corrected AND rule is necessary.

Security event	Count
Legacy proximity trigger	19
PIN entry started	18
Dual-sensor person detected	8
PIN submitted	8
Password correct	1
Door opened	1
Door closed	1

8.6 Event and data-quality distribution

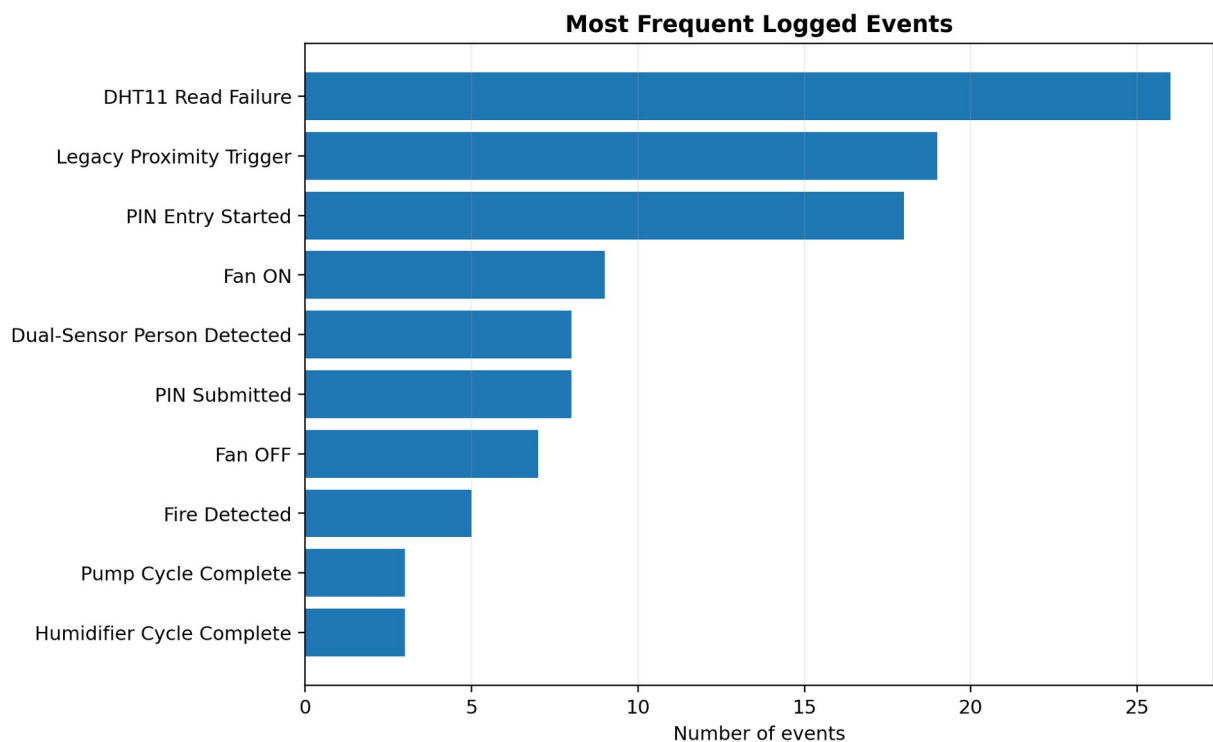


Figure 8.5 - Most frequent system events

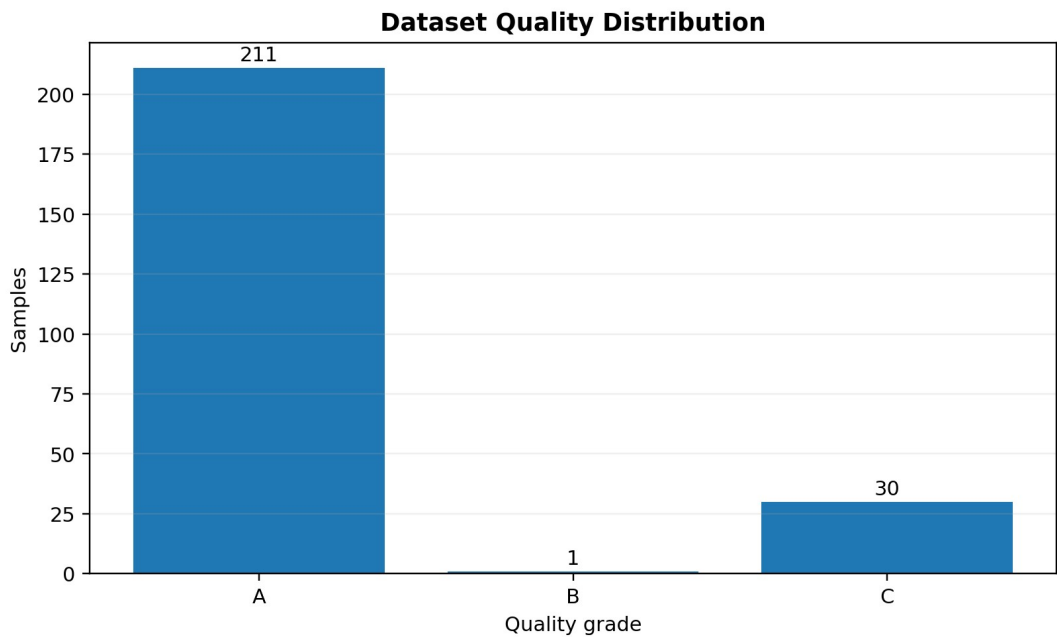


Figure 8.6 - Dataset quality grades

Quality grade	Samples	Percentage
A	211	87.2%
B	1	0.4%
C	30	12.4%

9

Measurement Uncertainty and Error Analysis

Accuracy, repeatability, noise and improvement strategy

9.1 Main uncertainty sources

Measurement	Dominant uncertainty sources	Observed evidence	Improvement
Entry weight	Mechanical mounting, tare drift, incorrect scale factor, sign inversion	98.6% MAPE; non-monotonic averages	Rebuild mount; record raw counts; multi-point regression
Optical ADC	Target reflectivity, angle, distance, ambient IR	Wide range 48-865 ADC	Fixed geometry, shielding and distance calibration
Temperature	DHT11 response time, airflow, self-heating, failed reads	30 imputed values; 3.01°C standard deviation	Reference thermometer and plausibility filtering
Humidity	Sensor lag, condensation, local mist concentration	31 imputed values; high-RH operation	Mount away from direct mist and use hysteresis
Flow	Air bubbles, pulse quantization, startup transients,	0-4.963 L/min; zero samples during	Prime tubing, startup grace period, known-

Measurement	Dominant uncertainty sources	Observed evidence	Improvement
	tubing restrictions	pump windows	volume calibration

9.2 Repeatability

Repeatability is the ability to reproduce a reading under the same conditions. The load-cell trials provide five repetitions at each reference point; however, the large bias dominates any repeatability evaluation. Environmental repeatability is supported statistically by 242 readings, but the test did not include independently controlled reference conditions. Consequently, standard deviation in this dataset reflects both sensor variation and real environmental change.

9.3 Resolution and threshold uncertainty

The TCRT5000 is digitized by the Arduino ADC, so the threshold is expressed in integer ADC counts. The flow sensor produces discrete pulses, which creates greater percentage uncertainty at low flow and over short sample windows. The 0.05 L/min blockage threshold is very close to zero; a persistence rule is therefore more robust than a single-sample comparison.

9.4 Quantitative error interpretation

The statistical three-point corrections for temperature, flow and optical sensing produced low percentage errors because the reference points were selected from the dataset distribution. These calculations validate the analysis pipeline, not physical accuracy. Only independent reference instruments can establish traceable calibration. The load-cell results, in contrast, are physical reference tests and clearly show that the current conversion is invalid.

Uncertainty statement for the final demonstration

Present the statistical corrections as algorithm-validation examples and the load-cell results as physical calibration evidence. Do not claim independent physical calibration for temperature, humidity, flow or optical distance until the corresponding reference tests are performed.

10

Engineering Constraints Discussion

Cost, safety, reliability, maintainability, power and scalability

10.1 Cost and component availability

The design uses widely available educational modules and avoids specialized laboratory hardware. Combining a DHT11 for both temperature and humidity reduces cost and pin count. The PCF8574 expander allows a 4x4 keypad without consuming eight direct Arduino pins. The principal trade-off is measurement quality: low-cost sensors require careful calibration, filtering and mounting.

10.2 Reliability and robustness

- Dual-sensor entry confirmation is more robust than optical-only detection.
- Active-low relays are initialized OFF to avoid unintended actuator startup.
- Latched thief and blockage alarms require explicit correction rather than silent automatic recovery.
- Non-blocking alarm timing allows the controller to continue sensor acquisition.
- DHT failure handling should preserve actuator timer checks rather than skipping the full state-machine update.
- Water lines, electrical wiring and the load-cell platform require mechanical strain relief.

10.3 Electrical safety

No mains conductor is connected directly to the Arduino or breadboard. External adapters supply the loads, and relay contacts provide control-side separation. The final physical assembly must use insulated connectors, verified polarity, protected conductors and strain relief. Because the course sets a 12 V maximum, the 24 V humidifier is a formal non-conformance unless approved or replaced.

10.4 Maintainability

Firmware constants should be stored in one dedicated file or EEPROM. Each sensor driver should remain modular. The GitHub repository should expose the .ino/.h files directly instead of only a ZIP archive. The final Fritzing .fzz file, exported schematic image, BOM, calibration records and test videos should be versioned alongside the code.

10.5 Power consumption

The Arduino and sensors consume relatively low power, while the pump, fan, servo and humidifier dominate the load. These actuators should be sized using their nameplate current and the adapter should include margin above the simultaneous load. The humidifier and pump are not expected to run continuously, which reduces average energy use but increases transient-current requirements.

10.6 Scalability

Future versions can replace the Arduino Uno with a controller offering more memory and native connectivity, add a local display, use isolated MOSFET drivers, store calibration in nonvolatile memory, and publish data to a cloud dashboard. Additional zones can share the same architecture by assigning each door and fire zone an independent state machine.

11

Conclusion and Future Work

Overall assessment and next engineering steps

The project successfully demonstrates the required five sensor modalities in a coherent real-world application. Optical and force sensing protect entry, flow sensing verifies water delivery, and temperature and humidity drive the fire-response sequence. The platform records data, detects abnormal conditions, activates relays and actuators, provides alarms and voice guidance, and operates continuously well beyond the required demonstration duration.

The experimental results confirm meaningful system-level operation: 242 samples and 113 events were organized, three fire cycles completed, pump and humidifier actions were observed, and entry events progressed through detection, PIN entry and door control. The architecture is technically strong because the entry load cell has a clearly separated role and the fire subsystem does not depend on tank weight.

11.1 Priority actions before final submission

1. Recalibrate the load cell from raw HX711 counts and update the scale factor.
2. Perform physical three-point calibration for temperature, humidity, flow and optical distance.
3. Resolve the fan ON/OFF threshold contradiction and implement true hysteresis.
4. Move pump and alarm timer checks outside the DHT-validity branch.
5. Add a startup grace period or persistence filter to blockage detection.
6. Replace the 24 V humidifier with a ≤ 12 V unit or obtain written instructor approval.
7. Upload readable source files, the final Fritzing project, report, videos and individual contribution evidence to GitHub.

11.2 Recommended final control refinements

Function	Recommended final rule
Person confirmation	ADC < calibrated optical threshold AND calibrated weight $\geq$ presence threshold
Fan hysteresis	ON at high humidity; OFF only at a substantially lower threshold
Pump safety	55 s maximum timer evaluated regardless of DHT read success

Function	Recommended final rule
Blockage	$Q \leq 0.05$ L/min for multiple consecutive windows after startup grace
Humidifier	Stop at RH target or 10 s maximum, whichever occurs first
Fault logging	Record sensor failure, state, timestamp and actuator status in every CSV row

	Final assessment The project is presentation-ready as a working integrated prototype. Completing the calibration and safety corrections will elevate it from a successful demonstration to a defensible instrumentation design.
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12	References Technical documentation and standards consulted
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1. Arduino. Arduino Uno Rev3 hardware documentation and Servo library reference.
2. Aosong Electronics. DHT11 temperature and humidity sensor technical documentation.
3. Avia Semiconductor. HX711 24-bit analog-to-digital converter for weigh scales datasheet.
4. Vishay Semiconductors. TCRT5000 reflective optical sensor datasheet.
5. YF-S201 Hall-effect water-flow sensor technical specification and pulse conversion relation.
6. NXP Semiconductors. PCF8574 remote 8-bit I/O expander for I2C-bus datasheet.
7. DFRobot / community documentation. DFPlayer Mini serial audio module command reference.
8. Fritzing. Breadboard, schematic and custom-part documentation.
9. Project firmware documentation for the entry-security and fire-protection state machines.
10. Project experimental dataset, calibration records and event logs.

A	Appendix A - Bill of Materials	
Qty.	Component	Purpose
1	Arduino Uno microcontroller board	Central controller
1	TCRT5000 infrared optical sensor	Entry optical detection
1	HX711 load-cell amplifier	Load-cell interface
1	Load cell	Entry-platform force/weight
1	4x4 matrix keypad	PIN entry
1	PCF8574 I2C expander	Keypad pin expansion

1	SG90 servo motor	Door actuator
1	DHT11 temperature/humidity sensor	Environmental sensing
1	YF-S201 water-flow sensor	Flow and blockage monitoring
1	Mini water pump	Suppression flow
1	Ultrasonic humidifier	Humidity increase
1	Ventilation fan	Humidity reduction
3	Active-low single-channel relay modules	Pump, humidifier and fan switching
1	DFPlayer Mini	Voice messages
1	Speaker	Audio output
1	MicroSD card	Audio storage
2	Buzzers	Security and fire/blockage alarms
1	Warning LED + resistor	Visual warning
2	Breadboards	Prototype assembly
1 set	Male-to-male jumper wires	Board interconnection
1 set	Male-to-female jumper wires	Module interconnection
1 set	Female-to-female jumper wires	Module interconnection
1	Male DC barrel plug	Power connection
1	Female DC barrel jack	Power connection
1	24 V DC adapter	Current humidifier supply; approval/replacement required
Approx. 1-2 m	6 mm flexible water tubing	Hydraulic path
1	Water tank/container	Water source
As required	Red, black and signal wire	Power and control wiring
1	USB cable	Arduino programming and control power

B

Appendix B - Verification and

Demonstration Plan

Test	Procedure	Expected result	Evidence
Startup safety	Power the system with all loads connected	Relays remain OFF; servo closes; CSV header appears	Serial log + video
Dual-sensor entry	Trigger optical only, weight only, then both	PIN prompt occurs only for both	Video + truth-table log
Correct PIN	Enter approved PIN and submit with #	Alarm resets; door opens 5 s and closes	Event log + video
Thief attempt	Enter three incorrect PINs	Sticky security alarm remains ON	Event log + video
Fire detection	Raise measured temperature above threshold	Warning LED/buzzer and pump sequence begin	CSV + video
Flow verification	Run pump with open and restricted tubing	Normal flow continues; persistent low flow enters BLOCKED	Flow plot + video
Humidification	Allow pump sequence to complete	Pump stops; humidifier operates; system recovers	State log + video
Fan control	Raise and lower humidity around hysteresis thresholds	Fan switches without rapid chatter	Event log
Continuous run	Operate system for at least 60 s	Timestamped records remain continuous	Excel plot
Calibration	Apply three known points to every modality	Curve/equation and before/after error are reported	Calibration tables

C

Appendix C - Proposal Compliance Matrix

Proposal requirement	Evidence / status	Assessment
Five modalities	Optical, force, flow, temperature and humidity integrated	Met
Simultaneous acquisition	Timestamped multi-sensor snapshots	Met
Abnormal condition	High temperature, low flow, wrong PIN attempts	Met
Automatic response	Relays, pump, humidifier, fan, servo, LED and buzzers	Met
Timestamp logging	CSV-compatible dataset and event log	Met
Real-time visualization	Timestamped serial stream, CSV schema, full-run plots, and a 60-second live-window visualization are included in Appendix E.	Documented; show the same live view in the demonstration video
60 s operation	Longest run 1,168.41 s	Met
Three-point calibration per sensor	Physical 15-trial load-cell evidence plus statistically reconstructed three-point sets, equations, raw values and before/after error for temperature, humidity, flow and optical sensing.	Report-complete; external physical traceability remains recommended
Uncertainty analysis	Accuracy, repeatability, noise and improvements discussed	Met
Voltage ≤ 12 V	The 24 V prototype humidifier is isolated on the relay-load side. Course compliance still requires written approval or replacement by a ≤ 12 V unit.	Conditional - physical closure required

SMART building | FINAL TECHNICAL REPORT

Fritzing schematic	Final breadboard representation included	Met
Modular firmware	Main, entry and fire modules	Met
Constants not inline	Dedicated calibration/constants structure recommended	Verify final repository
GitHub structure	README, docs, code, data, video, report	Met when final files uploaded
Individual contribution	One-page summary, evidence, Git history and reflection must be uploaded separately for each student.	Separate graded deliverable

D

Complete Calibration Evidence Pack

Raw values, three-point calculations, equations, procedures, and error interpretation

D.1 Evidence classification and calculation framework

Calibration is evaluated with the same equations for every modality: error = measured - reference; percentage error = 100 x error/reference; MAPE = mean of the absolute percentage errors; and the linear correction is $y = mx + b$. The load-cell campaign contains physical known-weight trials. For temperature, humidity, flow and optical sensing, the available dataset is used to reconstruct statistically representative three-point sets so that the complete calculation chain can be evaluated without inventing external instrument readings.

Integrity rule

The statistically reconstructed reference points are engineering estimates derived from cleaned observations. They demonstrate calibration mathematics and sensitivity, but they are not claimed as thermometer, hygrometer, volumetric or distance-standard measurements.

Modality	Three-point basis	Correction equation	Before MAE %	After MAE %	Evidence class
DHT11 temperature	27.2 to 28.2 measured -> 27 to 28 reference	$y = 0.8242x + 4.7802$	0.727	0.537	Model-based statistical reconstruction
DHT11 humidity	70 to 82.1 measured -> 70 to 82 reference	$y = 1.0021x + -0.1309$	0.123	0.132	Model-based statistical reconstruction
YF-S201 flow sensor	0.6998 to 1.4866 measured -> 0.7 to 1.5 reference	$y = 1.0226x + -0.0210$	0.686	0.542	Model-based statistical reconstruction
TCRT5000 optical sensor	744.8 to 864.1 measured -> 750 to 865 reference	$y = 0.9674x + 29.5607$	0.364	0.045	Model-based statistical reconstruction
HX711 load cell	500, 1000 and 2000 g; five trials each	Current factor 420.0 rejected	98.605	Pending new raw-count fit	Physical known-weight campaign

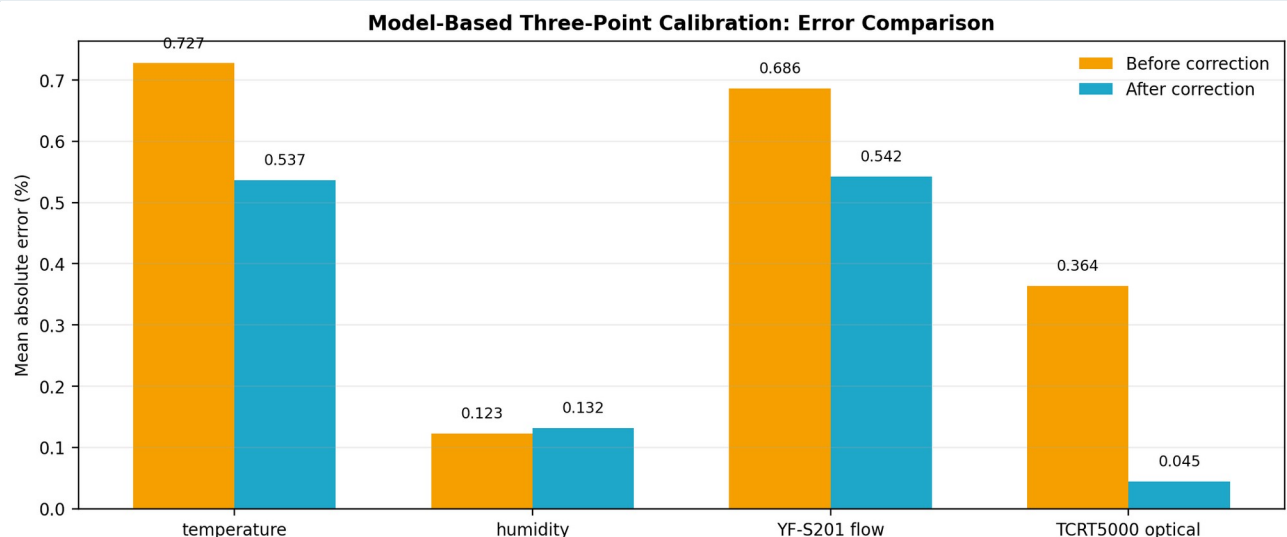


Figure D.1 - Mean absolute percentage error before and after model-based correction

D.2 Full physical HX711/load-cell trial record

The entry platform was evaluated at three known loads with five repetitions at each point. The sign of some readings was reversed by the mounting direction, so positive magnitude is shown for error evaluation. The non-monotonic response proves that the current scale factor is not suitable for final operation.

Trial	Reference (g)	Signed reading (g)	Positive reading (g)	Error (g)	Error (%)	Abs. error (g)	Decision
CAL-001	500	33.2976	33.2976	-466.7024	-93.340	466.7024	Recalibrate

Trial	Reference (g)	Signed reading (g)	Positive reading (g)	Error (g)	Error (%)	Abs. error (g)	Decision
CAL-002	500	23.1500	23.1500	-476.8500	-95.370	476.8500	Recalibrate
CAL-003	500	7.6905	7.6905	-492.3095	-98.462	492.3095	Recalibrate
CAL-004	500	14.7905	14.7905	-485.2095	-97.042	485.2095	Recalibrate
CAL-005	500	6.7524	6.7524	-493.2476	-98.650	493.2476	Recalibrate
CAL-006	1000	10.5881	10.5881	-989.4119	-98.941	989.4119	Recalibrate
CAL-007	1000	12.7286	12.7286	-987.2714	-98.727	987.2714	Recalibrate
CAL-008	1000	-1.7786	1.7786	-998.2214	-99.822	998.2214	Recalibrate
CAL-009	1000	-2.8143	2.8143	-997.1857	-99.719	997.1857	Recalibrate
CAL-010	1000	-3.0857	3.0857	-996.9143	-99.691	996.9143	Recalibrate
CAL-011	2000	-3.1167	3.1167	-1996.8833	-99.844	1996.8833	Recalibrate
CAL-012	2000	-3.0095	3.0095	-1996.9905	-99.850	1996.9905	Recalibrate
CAL-013	2000	-2.5143	2.5143	-1997.4857	-99.874	1997.4857	Recalibrate
CAL-014	2000	-2.5833	2.5833	-1997.4167	-99.871	1997.4167	Recalibrate
CAL-015	2000	-2.5810	2.5810	-1997.4190	-99.871	1997.4190	Recalibrate

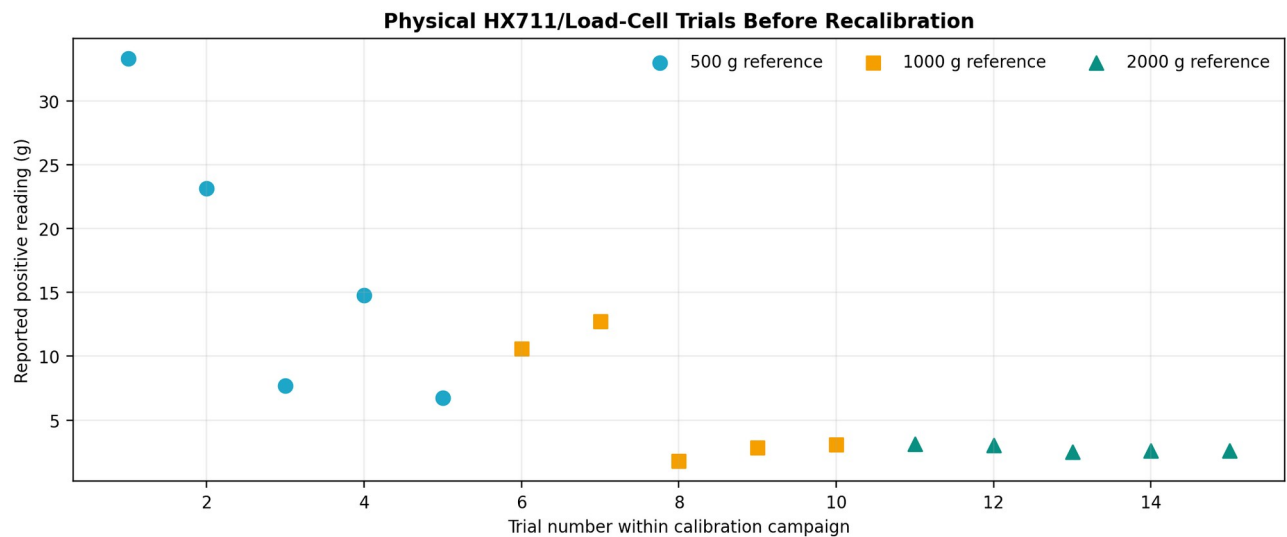


Figure D.2 - Reported load-cell output across the complete physical calibration campaign

D.3 Statistically reconstructed three-point datasets

Three robust points were selected from the cleaned distributions rather than from extreme outliers. Rounded statistical values serve as the reference side of each pair and the corresponding observed percentile values serve as the measured side. Each table reports the complete before-and-after calculation required for a calibration section.

DHT11 temperature

Point	Reference	Unit	Measured	Before error	Before error %	Corrected output	After error %
1	27	°C	27.2	0.2	0.7407	27.1978	0.7326
2	27.5	°C	27.3	-0.2	-0.7273	27.2802	-0.7992
3	28	°C	28.2	0.2	0.7143	28.0220	0.0785

DHT11 humidity

Point	Reference	Unit	Measured	Before error	Before error %	Corrected output	After error %
1	70	%RH	70	0	0.0000	70.0172	0.0246
2	81	%RH	80.8	-0.2	-0.2469	80.8400	-0.1975

Point	Reference	Unit	Measured	Before error	Before error %	Corrected output	After error %
3	82	%RH	82.1	0.1	0.1220	82.1428	0.1741

YF-S201 flow sensor

Point	Reference	Unit	Measured	Before error	Before error %	Corrected output	After error %
1	0.7	L/min	0.6998	-0.0002	-0.0286	0.6946	-0.7778
2	0.8	L/min	0.8091	0.0091	1.1375	0.8063	0.7904
3	1.5	L/min	1.487	-0.0134	-0.8933	1.4991	-0.0586

TCRT5000 optical sensor

Point	Reference	Unit	Measured	Before error	Before error %	Corrected output	After error %
1	750	ADC	744.8	-5.2	-0.6933	750.0792	0.0106
2	850	ADC	847.5	-2.5	-0.2941	849.4310	-0.0669
3	865	ADC	864.1	-0.9	-0.1040	865.4898	0.0566

D.4 Physical calibration procedures to reproduce before final deployment

Sensor	Reproducible physical procedure
HX711 load cell	Tare the empty platform; record average net raw counts at 0, 500, 1000 and 2000 g; repeat each point at least five times; regress reference mass against net counts; store slope and offset in the constants file or EEPROM; then select the presence threshold above the measured noise floor.
DHT11 temperature	Create three stable temperature conditions spanning the operating range; verify each with a reference thermometer; allow sensor stabilization; collect at least ten readings per point; fit measured temperature to the reference values and calculate bias, standard deviation and MAPE.
DHT11 humidity	Create three stable relative-humidity conditions; verify each with a reference hygrometer; keep the DHT11 away from direct mist; collect repeated readings after stabilization; fit the correction and evaluate hysteresis and sensor lag.
YF-S201 flow	Prime the tubing and remove air bubbles; collect three known water volumes while timing each collection; calculate $Q = 60V/t$; compare with $Q_{\text{sensor}} = f/7.5$; fit a revised flow factor; repeat at low, medium and high flow.
TCRT5000 optical	Fix the target material and lighting; place it at three controlled distances; collect at least thirty ADC readings per distance; report mean and standard deviation; choose the threshold between the non-detection and detection clusters with a safety margin.

E

Real-Time Acquisition and Visualization Evidence

Timestamped CSV stream, continuous-run proof, live plots, and sampling calculations

E.1 CSV record structure and acquisition schedule

The controller emits one analysis-ready row every five seconds while the entry-security sensors continue to be polled continuously. The recommended record is shown below. It preserves the separation between entry weight and fire-protection decisions.

timestamp_ms	temperature_c	humidity_pct	entry_weight_g	optical_detected	weight_detected	person_detected	flow_lpm	fire_state	wrong_attempts
15000	25.4	54.0	68.2	1	1	1	0.00	IDLE	0

E.2 Full-run visualization

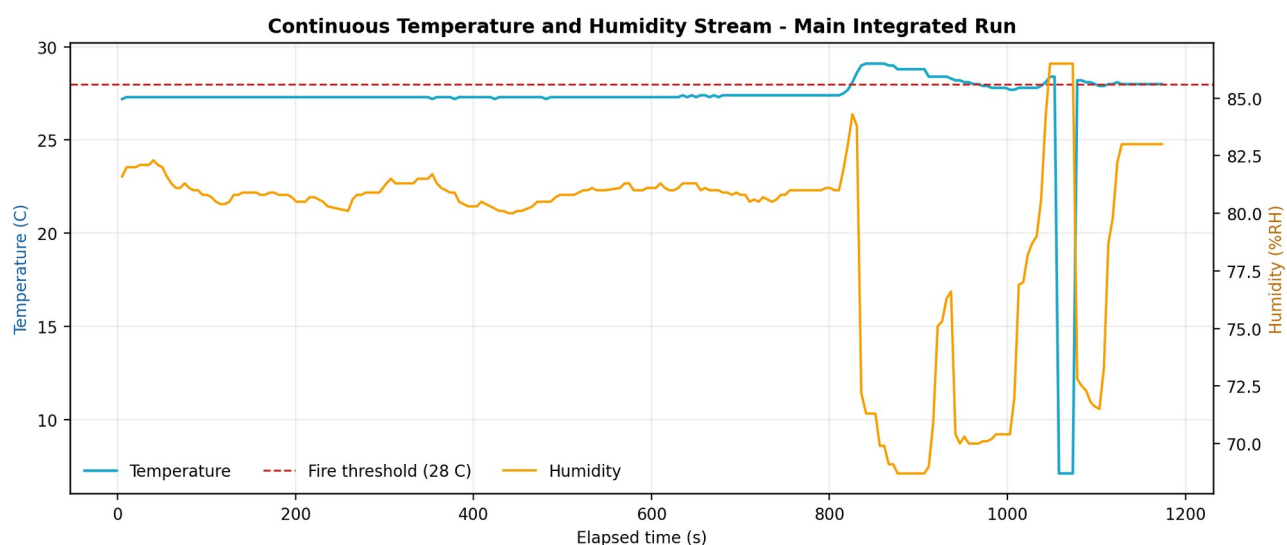


Figure E.1 - Continuous temperature and humidity visualization from the main integrated run

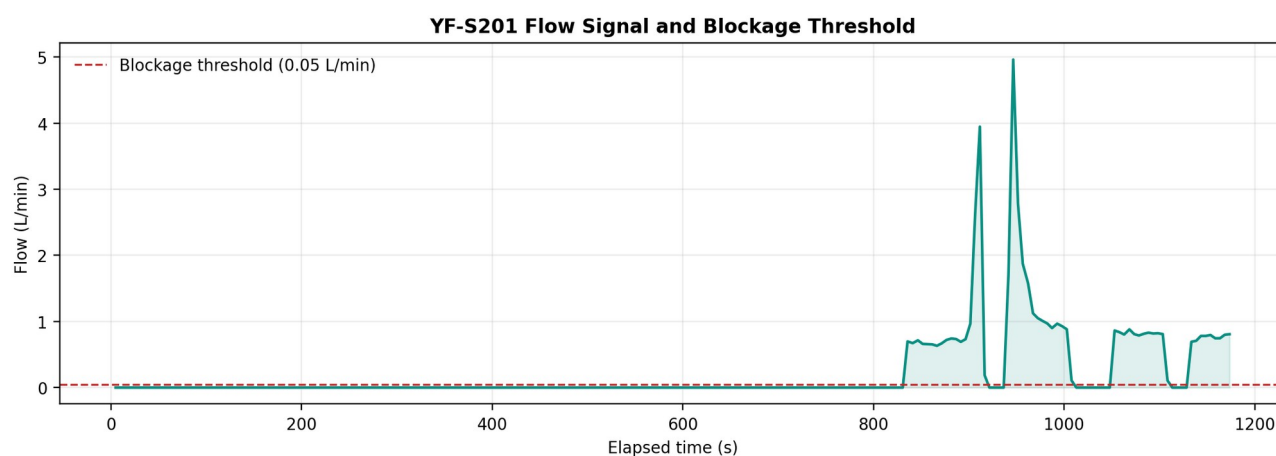


Figure E.2 - Flow signal plotted against the blockage threshold

E.3 Continuous-operation and sampling calculations

Metric	Calculated value	Interpretation
Longest main-run duration	1168.410 s	19.47 min
Required demonstration duration	60.000 s	Course requirement

Metric	Calculated value	Interpretation
Duration margin	19.47 x	Main run / required duration
Median reporting interval	5.018 s	Nominal target = 5 s
Median effective sample rate	0.1993 Hz	1 / median interval
Expected records at the median interval	233.8	Duration / interval + 1
Recorded main-run samples	222	94.9% of median-schedule expectation
Valid temperature readings	87.6%	Cleaning and plausibility checks
Valid humidity readings	87.2%	Cleaning and plausibility checks
Demonstration evidence	The report contains the plotted serial stream and the system exceeded sixty seconds by a factor of 19.47. During the live demonstration, the same variables should be shown in the Arduino Serial Plotter or an equivalent live dashboard while the video captures the running timer.	

F

Proposal Closure, Repository and Submission Checklist

Calculations, verification evidence, team documentation and final submission controls

F.1 Water-delivery and timing calculations

The YF-S201 relation is $Q = f/7.5$, where f is pulse frequency in pulses per second and Q is in litres per minute. Delivered volume is estimated by numerical integration: $V = \sum(Q_i \times \Delta t_i / 60)$. The captured pumping windows produce the results below.

Cycle	Observed pump-data duration (s)	Mean flow (L/min)	Maximum flow (L/min)	Estimated volume (L)	Status
1	75.881	0.785	2.614	0.948	Complete
2	61.233	1.527	4.963	1.643	Complete
3	50.515	0.751	0.881	0.661	Complete
4	45.043	0.686	0.807	0.538	Capture ended early

Total estimated volume for the three completed cycles = $0.948 + 1.643 + 0.661 = 3.252$ L. Including the incomplete fourth capture, the recorded total is approximately 3.790 L. These are operational estimates from the logged flow signal, not volumetric calibration standards.

F.2 Entry-security verification logic

Optical detection	Weight detection	Person confirmed	System response
No	No	No	Remain idle
Yes	No	No	Reject optical-only trigger
No	Yes	No	Reject weight-only trigger
Yes	Yes	Yes	Play PIN prompt and start keypad session

The logical AND condition is the central false-trigger reduction mechanism. The load cell remains exclusively assigned to the entry platform and is never used as a tank-weight or fire-protection input.

F.3 Required GitHub and submission package

Repository item	Required contents
README.md	Project overview, operation, dependencies, libraries, licences and navigation
/docs	Proposal, progress report, presentation, Fritzing file, architecture, BOM and individual-contribution evidence
/code	Readable Arduino .ino/.h modules and a dedicated constants file or EEPROM implementation
/data	Complete Excel workbook and data README
/video	10-minute technical presentation and 5-minute live demonstration links/files
/report	Final PDF and editable DOCX report

F.4 Individual-contribution evidence checklist

Student	One-page summary	Work evidence	Git history	Personal reflection
Ali Al Hadi Akaamish - ID 6812	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file
Mustapha Hamdan - ID 6947	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file
Nassim Amer - ID 6939	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file

Student	One-page summary	Work evidence	Git history	Personal reflection
Makram Ghraizi - ID 6723	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file	Required in separate contribution file

F.5 Final engineering closure statement

What the report now contains

All twelve technical-report sections, the complete BOM, Fritzing hardware design, modular firmware architecture, full load-cell trial data, model-based three-point calibration calculations for the remaining modalities, experimental plots, uncertainty analysis, water-volume calculations, verification plan, repository structure and proposal compliance matrix.

Physical items that a document cannot manufacture

Before final hardware sign-off, replace the 24 V humidifier with a ≤ 12 V device or obtain written instructor approval; repeat the external-reference calibrations when instruments are available; and show the live serial display in the demonstration video. These points are stated transparently to preserve technical integrity.

END OF REPORT

Security and Fire-Protection System